GenAI Tools Overview: Uses and Nuances

UQ Library

Acknowledgement of Country

The University of Queensland (UQ) acknowledges the Traditional Owners and their custodianship of the lands on which we meet.

We pay our respects to their Ancestors and their descendants, who continue cultural and spiritual connections to Country.

We recognise their valuable contributions to Australian and global society.



The Brisbane River pattern from *A Guidance Through Time* by Casey Coolwell and Kyra Mancktelow.

Contents



01

Prompting Strategies



02

Literature Searching



03

Ethics & Integrity



04

Other tools of note

Generative AI

Generative AI is AI that can produce an output based on a text prompt.

Outputs can be:

- Text
- Images
- Code
- Multimedia

Asking Bing Copilot what Generative AI is tells us:
Generative AI, sometimes referred to as gen AI, is a branch of artificial intelligence (AI) that can create original content in response to a user's prompt or request. This content can include text, images, video, audio, or even software code. Generative AI relies on sophisticated machine learning models, specifically deep learning models, which simulate the learning and decision-making processes of the human brain.

Popular Generative AI Tools

See Library's guide Artificial Intelligence – AI tools for students at UQ

Text (and more):

OpenAI **ChatGPT**

Microsoft **Copilot**

Google **Gemini**

Perplexity AI

Image:

OpenAI **DALL-E**

Midjourney

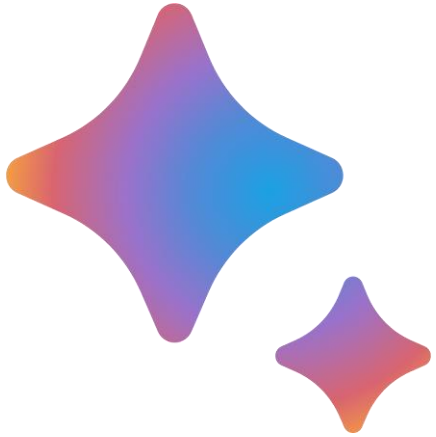
Stability AI **Stable Diffusion**

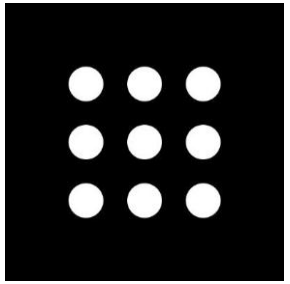
Audio:

OpenAI **Whisper**

Google **NotebookLM**









Cool things to do with AI…



AI and Tools for Expanding Access to Justice

The Cambridge Handbook of AI in Civil Dispute Resolution (forthcoming, Cambridge University Press)

17 Pages • Posted: 28 Jun 2024

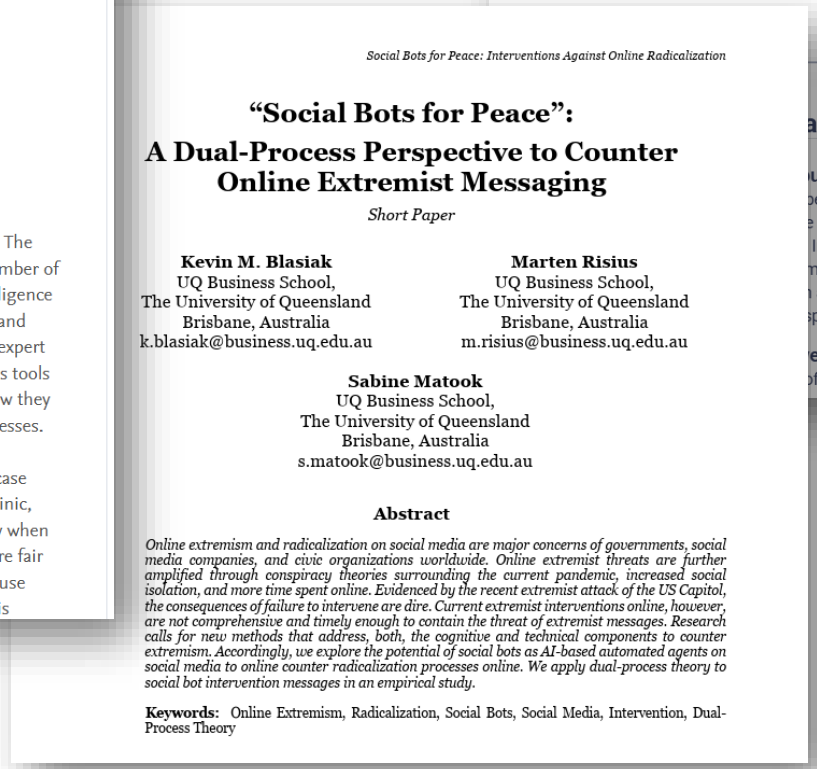
Quinten Steenhuis
Suffolk University Law School

Date Written: June 30, 2024

Abstract

Access to justice is a fundamental right, yet for millions of people around the world resolution of legal problems via both formal and informal means isn't available, affordable, accessible, or understandable. The high cost of legal help, now between $200 and $400/hour across the United States, and the limited number of attorneys providing free legal aid, is just one of these barriers. This chapter explores how artificial intelligence (AI) is making a practical difference in the problems of everyday people by automating legal processes and making legal systems more affordable and accessible. This chapter discusses generative AI, traditional expert systems, and computational approaches each as tools that can help legal aid and private lawyers, and as tools that can help unrepresented people. Each section demonstrates specific tools in action, showcasing how they reduce the need for direct legal intervention while maintaining the accuracy and integrity of legal processes. Integrating AI into legal practices isn't without its challenges. Traditional rules-based systems can be inflexible, and statistical systems can introduce issues such as algorithmic bias. Through the lens of 3 case studies: Massachusetts Defense for Eviction, The Resurrection Project's Migrant Work Authorization Clinic, and Rentervention, we will discuss the need for human oversight and co-authoring with human review when using generative AI. This part of the chapter will highlight the importance of developing AI tools that are fair and transparent to ensure they help rather than hinder access to justice. We will discuss inappropriate use cases for AI and ways to mitigate risk when AI is employed in Access to Justice solutions. Ultimately, this

Steenhuis, Q. (forthcoming). AI and Tools for Expanding Access to Justice. In *The Cambridge Handbook of AI in Civil Dispute Resolution*. Cambridge University Press.



Social Bots for Peace: Interventions Against Online Radicalization

"Social Bots for Peace":
A Dual-Process Perspective to Counter Online Extremist Messaging

Short Paper

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Abstract

Online extremism and radicalization on social media are major concerns of governments, social media companies, and civic organizations worldwide. Online extremist threats are further amplified through conspiracy theories surrounding the current pandemic, increased social isolation, and more time spent online. Evidenced by the recent extremist attack of the US Capitol, the consequences of failure to intervene are dire. Current extremist interventions online, however, are not comprehensive and timely enough to contain the threat of extremist messages. Research calls for new methods that address, both, the cognitive and technical components to counter extremism. Accordingly, we explore the potential of social bots as AI-based automated agents on social media to online counter radicalization processes online. We apply dual-process theory to social bot intervention messages in an empirical study.

Keywords: Online Extremism, Radicalization, Social Bots, Social Media, Intervention, Dual-Process Theory

Blasiak, K. M. et al. (2021). "Social Bots for Peace": A Dual-Process Perspective to Counter Online Extremist Messaging. *42nd International Conference on Information Systems, ICIS 2021 TREOs*.



Published on 24.10.2024 in Vol 7 (2024)

Preprints (earlier versions) of this paper are available at https://preprints.jmir.org/preprint/54210, first published November 02, 2023.

Developing Independent Living Support for Older Adults Using Internet of Things and AI-Based Systems: Co-Design Study

Claire M Timon[1]; Emma Heffernan[2]; Sophia Kilcullen[3]; Louise Hopper[4]; Hyowon Lee[5]; Pamela Gallagher[4]; Alan F Smeaton[6]; Kieran Moran[7]; Pamela Hussey[8]; Catriona Murphy[2]

Cited by (1) Tweetations (1) Metrics

act

und:
er of older people with unmet health care and support needs is increasing substantially
challenges facing health care systems worldwide. There are potentially great benefits to
Internet of Things coupled with artificial intelligence to support independent living and the
ment of health risks, thus improving quality of life for the older adult population. Taking a
approach has the potential to ensure that these technological solutions are developed to
specific user needs and requirements.

e:
f this study was to investigate stakeholders' perceptions of independent living and

Timon C. et al. (2024). Developing Independent Living Support for Older Adults Using Internet of Things and AI-Based Systems: Co-Design Study. *JMIR Aging*, e54210. DOI: 10.2196/54210

AI Tools & Research

AIs available or approved at UQ

MS Copilot

Grammarly

ChatGPT

Claude

Lexis Protégé

Phil AI (IBISWorld)

- AI chat bots using information scraped from the internet
  - In the case of Lexis Protégé; has been specifically built for legal research.
- Has access to some research information

AI tools from the library guide “Overview of AI tools for literature searching”

Elicit

Scite

SciSpace

Consensus

Keenious

OpenAlex

- Designed with “scholarly information” as a base
- Many use Semantic Scholar as source
- Relies on RAG (Retrieval Augmented Generation) for information

https://guides.library.uq.edu.au/how-to-find/quality-information/ai-tools-for-literature-searching

Prompting strategies

Assign a role

- Student
- Academic
- Marketing expert
- Business owner

Provide instructions

- Do Research
- Give me Vivid descriptions
- Advise me
- Help me Brainstorm

Add context

- Target audience or reading level
- Specify the background information relevant to the prompt

Apply constraints

- Define output (dotpoints, list, word limit etc)
- Currency of information e.g. find sources from 2025

Give examples

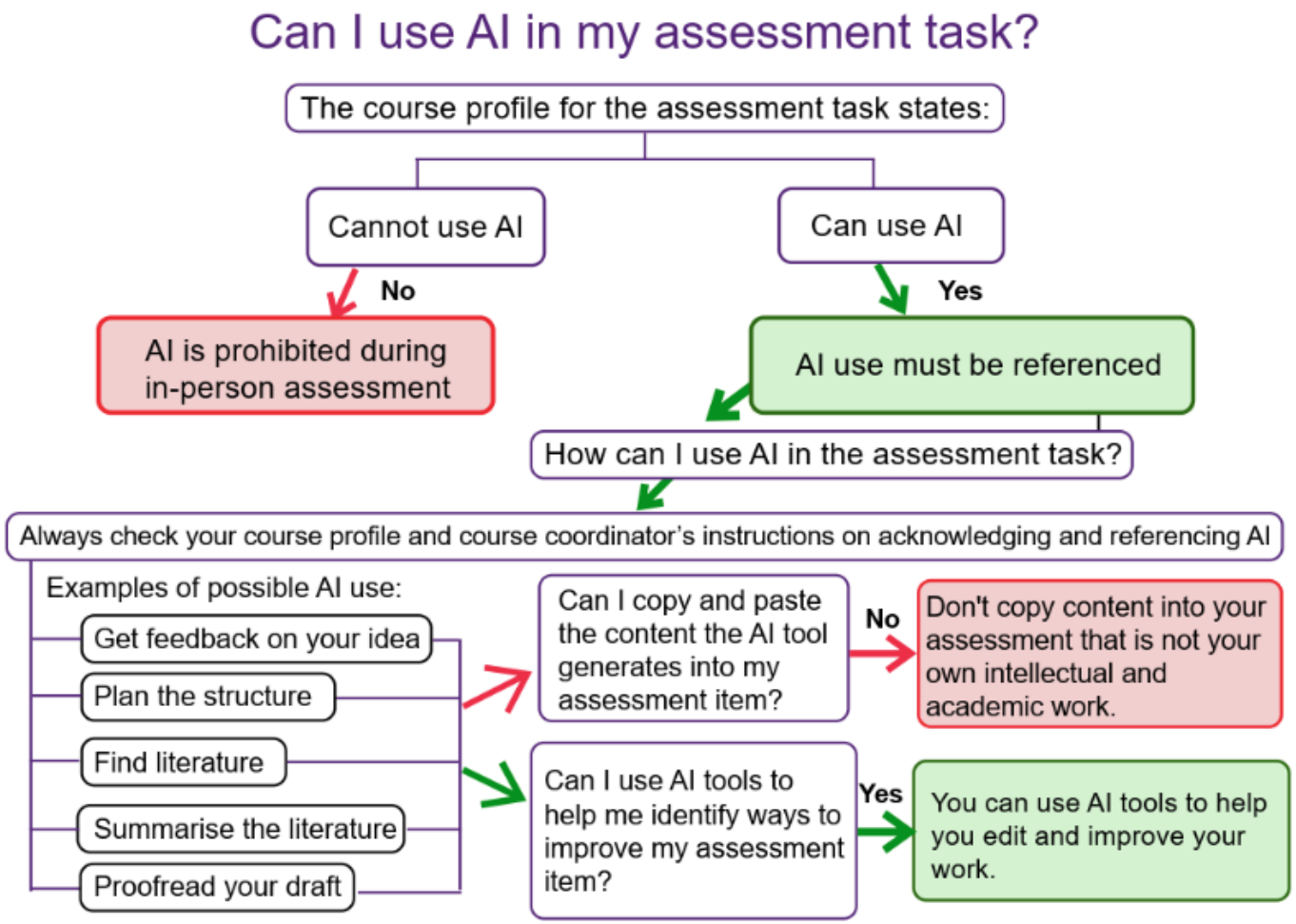
- Show it what is an acceptable answer
- Give it a starting point

AI and Academic Integrity

At UQ, the use of AI outputs without attribution, and contrary to any prohibition by teaching staff, is a form of plagiarism and constitutes academic misconduct. Check each assessment in your course profiles.

If you use AI in your assessment without permission or appropriate acknowledgment it may be considered misconduct. If you have questions, you should ask your course coordinator.

- Academic integrity and student conduct



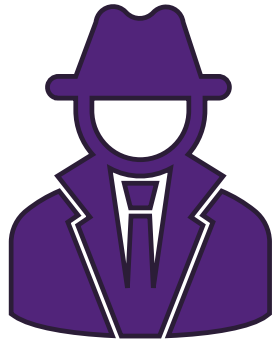
Acknowledge and reference AI use

- Acknowledge use of any AI tool (publishing form or cover sheet)
- Cite the tool and prompt
- Take note of what you used it for and keep copies of outputs
- If the AI output recommends sources, check these exist and cite these in your reference list if they inform the report, essay or document you are preparing
- Find more information:
  - Referencing Guide (Using AI in your Assignments)
  - Publishing Guide



Privacy

- AI tools store and learn from your inputs.
- Do not provide personal information to AI platforms.
- Check the privacy policy of the AI:
  - For example: OpenAI privacy policy.
- Do not upload UQ library licensed resources into an AI platform to summarise papers. This is not ethical or responsible behaviour:
  - see UQ Library Terms of Use.



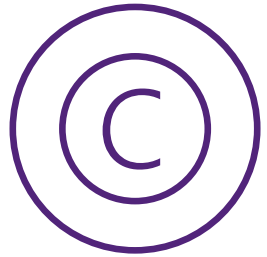
Copyright

Works created through Generative AI tools cannot be copyrighted

The Copyright Act 1968 (Cth) requires that the creator of a work is a qualified person. A qualified person is an Australian citizen or person resident in Australia.

Copyright Act 1968

- https://www.legislation.gov.au/Details/C2019C00042



Using AI Tools for Study & Research

THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

Basic LLMs/AI (MS CoPilot, ChatGPT, Claude)

- Generate ideas
- Explain topics

Research discovery based LLMs (Lexis Protégé, Phil AI, SciSpace, Elicit, Consensus)

- Used to discover papers, develop research topics – RAG based retrieval providing results as resource listings
- Chat with collections of papers
- Seed papers as a discovery tool – Research Rabbit, LitMaps

Developing uses

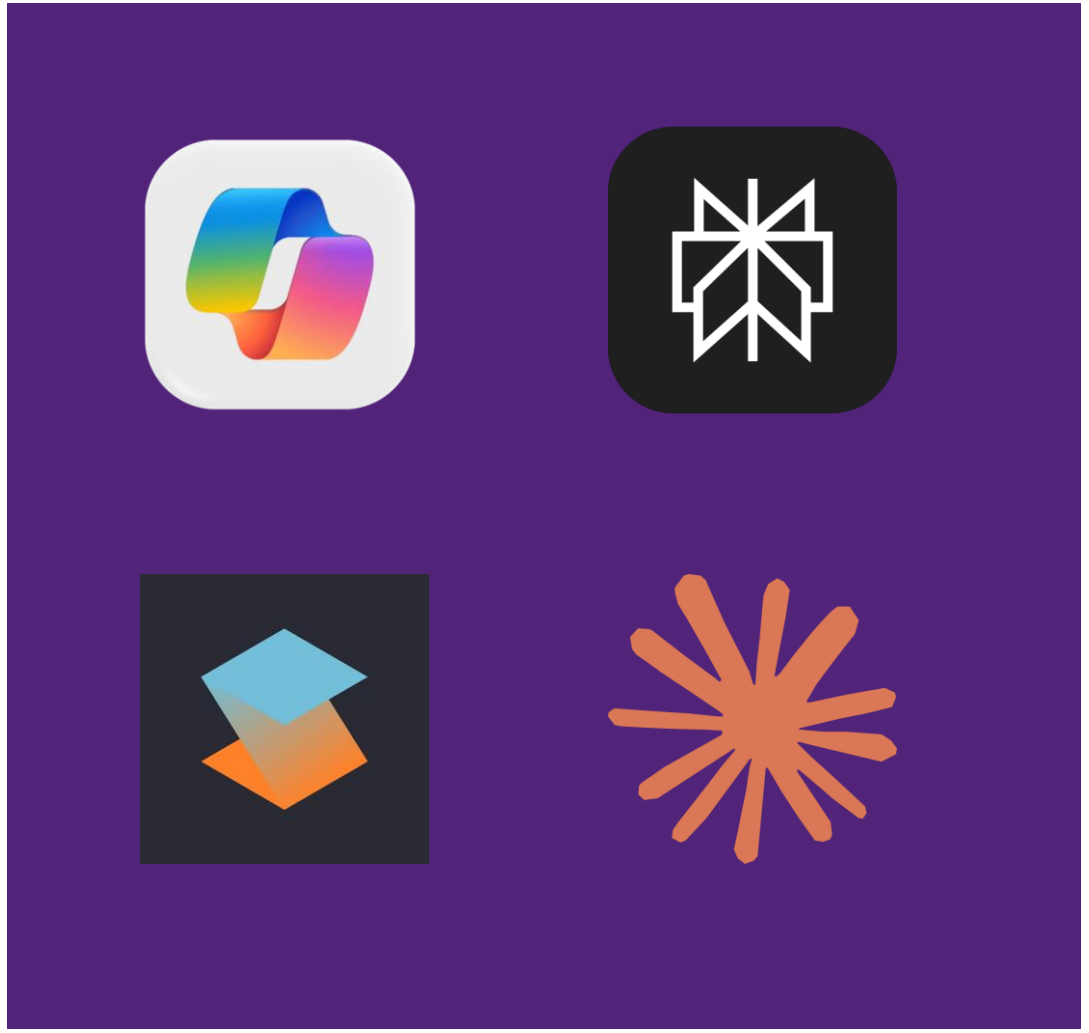
- Deep research tools (OpenAI, Gemini)
- Summarising (Lexis Protégé)

Demo/Play time

We encourage you to explore along with us

Using AI tools for:

- Literature searching
- Image analysis/interpretation
- Improving workflow
- Generating code



Improve Your Learning

Why learn something if AI can do it?

- AI gives answers – not understanding
- Overuse weakens critical thinking
- Learning builds judgement



You can use AI, but you still need to ***understand*** the output.

Improve Your Understanding

Share problems in stages

- Large problems can confuse both you and the AI

Give prompts step by step

- Break down the process, and ask for clarification as you go

Match solutions to your understanding

- Tell the AI what you know and ask where to start

Ask for tutorials from trusted sources

- AI can be a good teacher, but it can make things up



Improve Your Approach

Ask for help with direction

- How should I visualise this dataset?
- What tools are best for this task?

Ask for feedback on your current process

- Here is how I usually work. Is there a better way?
- Are there more efficient tools or steps?



*Only upload data you have permission to share.
Perhaps just share the headings/structure.

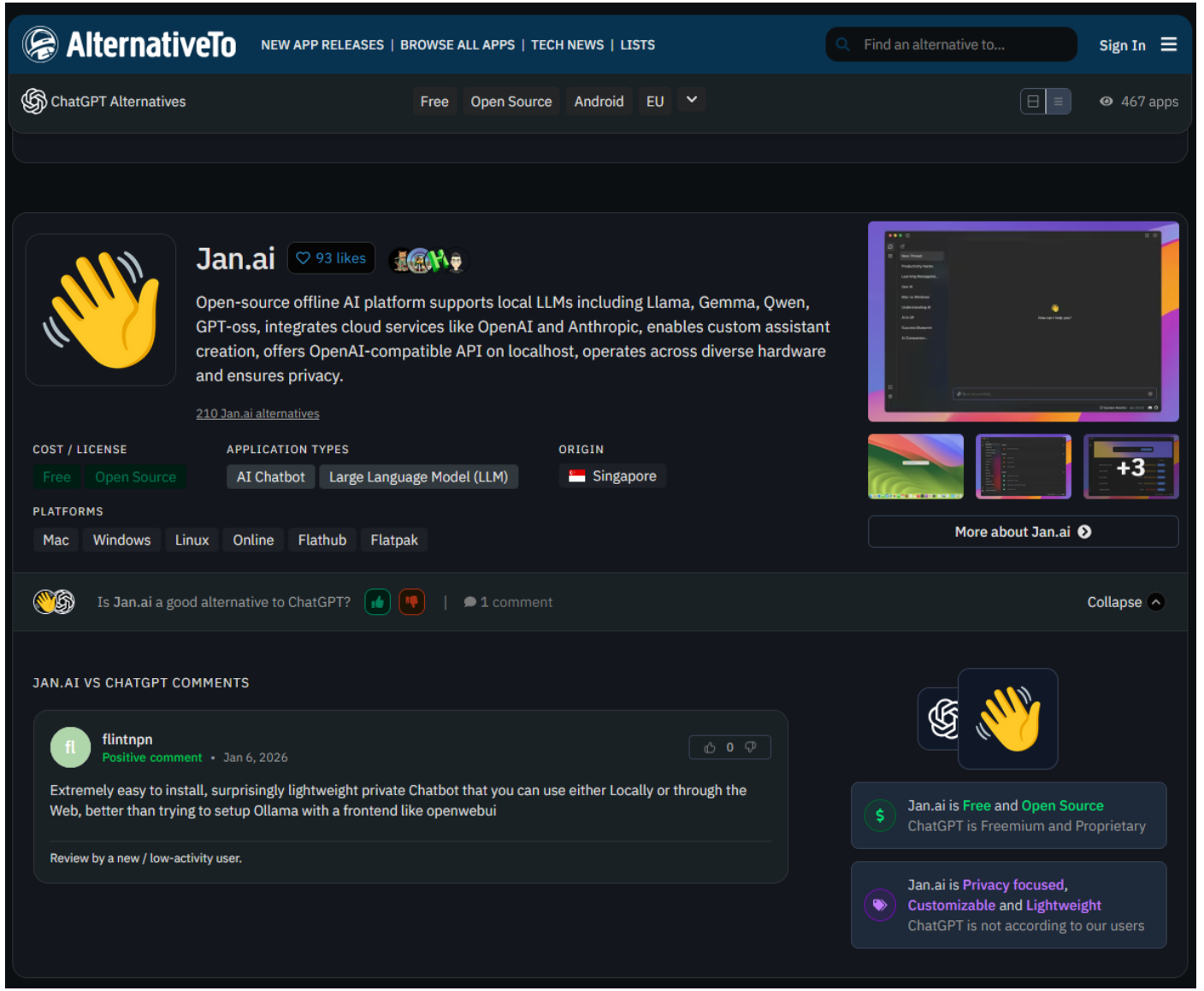
Finding the right tool

How to find alternatives?

Potential criteria:

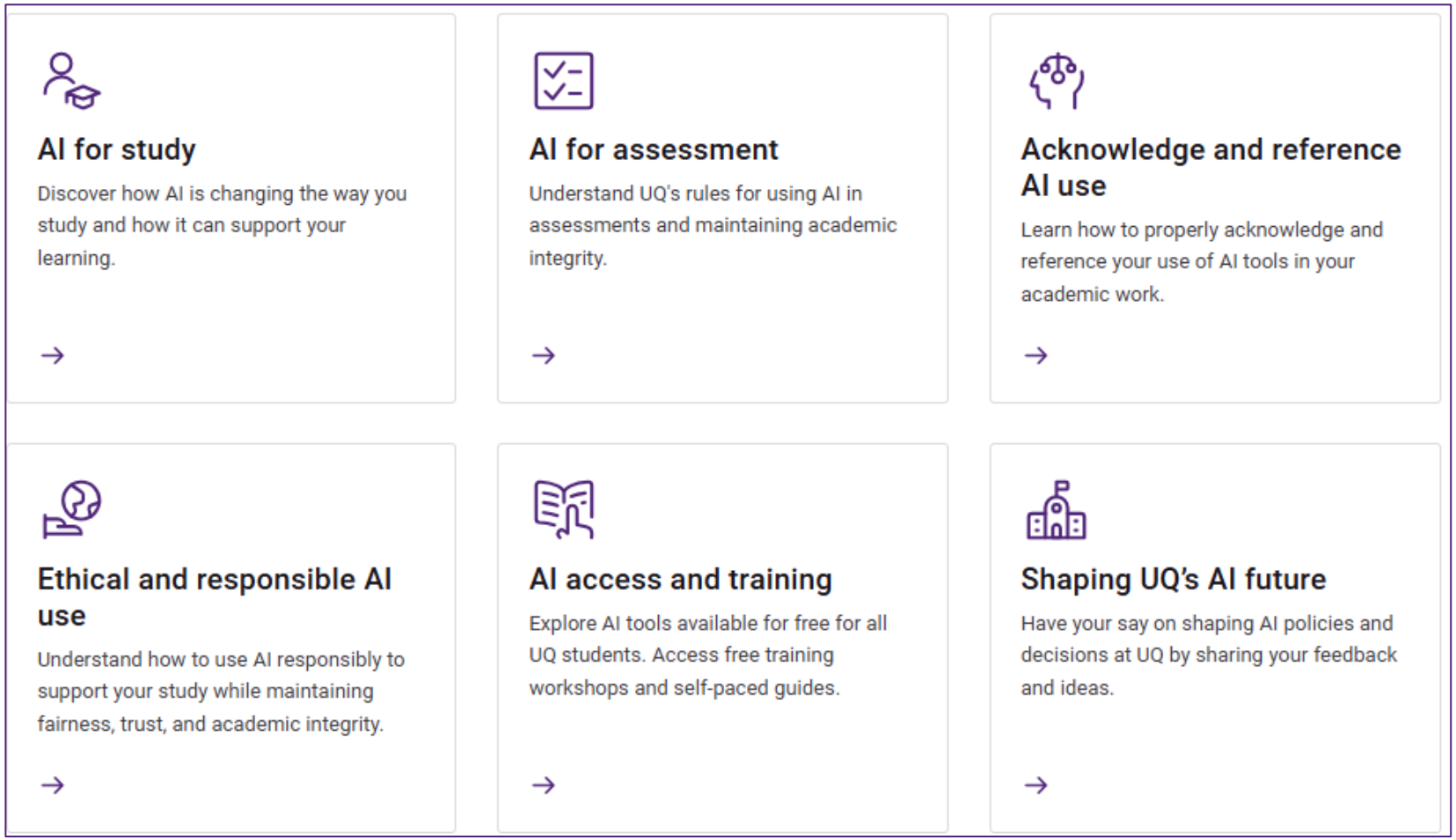
- Features?
- Licence?
- Cost?
- Privacy? Cloud or local?
- Who owns it?

… and do I even need AI in the first place?



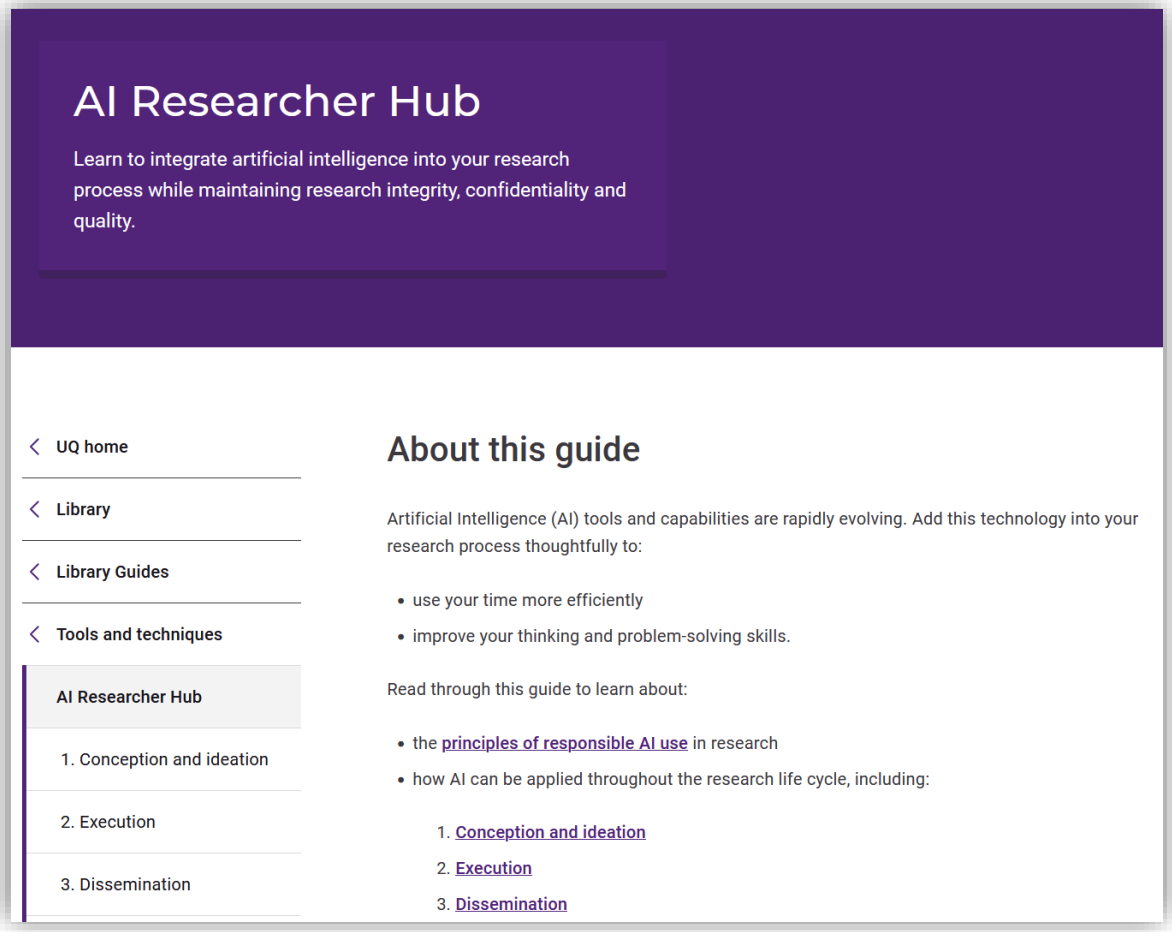
AI Student Hub

Learn how to use AI responsibly and confidently in your studies at UQ.



https://guides.library.uq.edu.au/tools-and-techniques/ai-student-hub

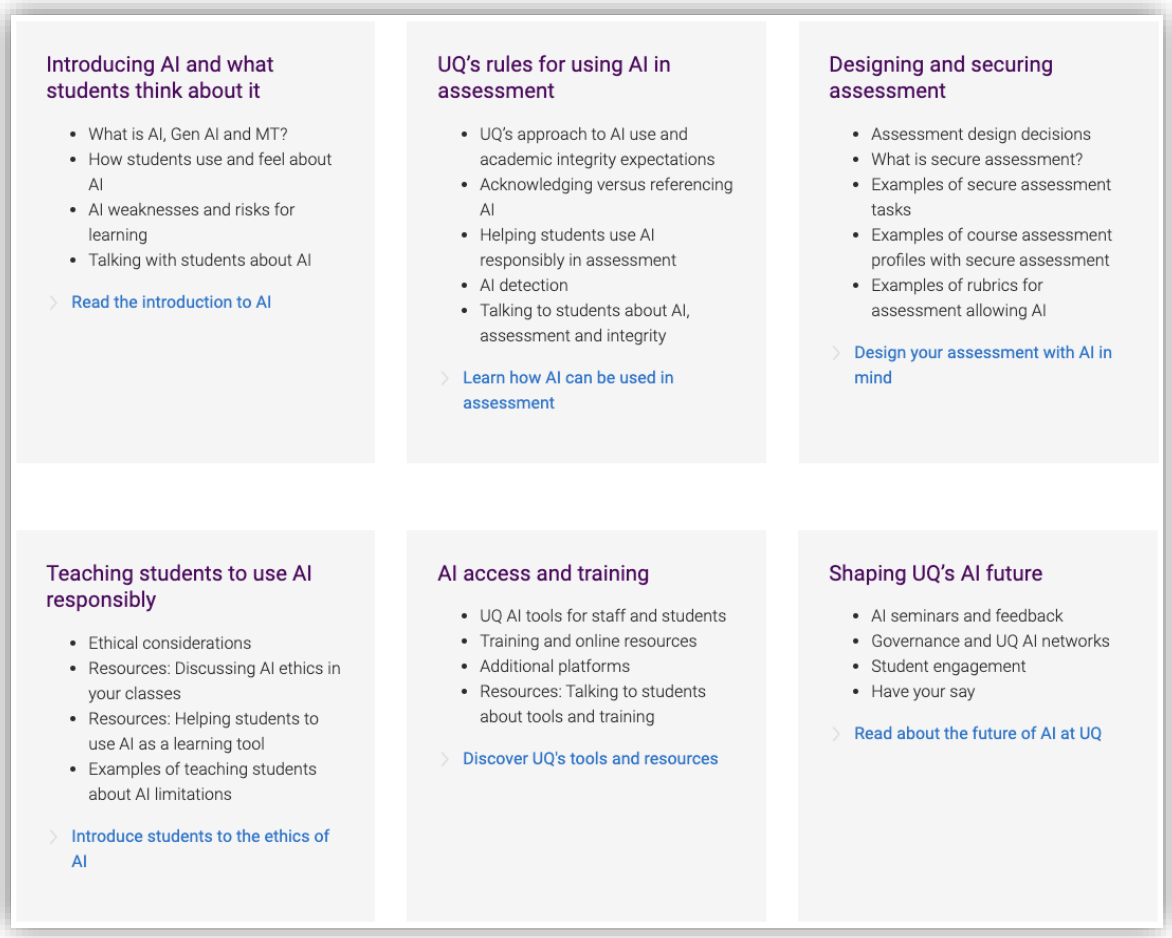
AI Researcher Hub



The Library has information and resources for researchers regarding using AI, centralised in the AI Researcher Hub.

Learn more:
https://guides.library.uq.edu.au/tools-and-techniques/ai-researcher-hub

AI Teacher Hub



UQ has information and resources for teaching staff regarding Generative AI, centralised in the AI Teacher Hub.

UQ provides **free access** to Microsoft Copilot and Adobe Creative cloud software including Firefly.

Learn more:

https://itali.uq.edu.au/teaching-guidance/ai-teacher-hub

Other resources of note

AI referencing guide

Artificial Intelligence Open-textbook

Contact

UQ Library

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Tech Trainers: training@library.uq.edu.au

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